Comprehensive Performance & Protocol Analysis Across Alternative Network Topologies — Extended Edition

Systematic Evaluation of TCP, UDP, HTTP, and DNS over Star, Mesh, Bus, and Hybrid Topologies Under Controlled Stress Conditions, with Extended Coverage of Security, QoS, Cost, Scalability, and Packet-Level Diagnostics.

Attribute	Detail
Course Outcomes	CO3 (Protocols), CO4 (Topology Design), CO5 (Metrics)
Simulation Platform	Cisco Packet Tracer / GNS3
Bloom's Taxonomy	Level 5 (Evaluate) & Level 6 (Create)
SDG Mapping	Goal 9 (Industry, Innovation & Infrastructure)
Network Scale	4 Interconnected Segments | 44 End Devices | 4 Core Topologies
Deliverables	Pseudocode, Topologies, Data Tables, CLI, Security & QoS Design, GitHub Setup

Modern campus and enterprise networks must support diverse traffic flows with contrasting Quality of Service (QoS) requirements. Real-time applications such as voice and video streaming demand low latency and minimal jitter, while transaction-oriented systems require guaranteed, zero-loss delivery. To rigorously assess how topology architecture influences protocol behaviour, this project investigates a standardized campus infrastructure comprising 4 distinct functional segments housing a total of 44 active end devices, exceeding the 40-device minimum threshold specified for the assignment.

1. Problem Understanding & Network Requirements

1.1 Traffic Characterization & Protocol Requirements

Protocol / Service	Transport & Port	Behavioural Profile
HTTP	TCP / Port 80	Web traffic from Academic and Administrative blocks to the Data Center. Bursty TCP sessions, three-way handshakes, slow-start window scaling, RTT-sensitive.
DNS	UDP / Port 53	Host-to-IP resolution. Short request/response packets, target latency < 20 ms, rapid timeout and retry, stateless transport.
VoIP / RTP Streaming	UDP / Port 5004	Continuous real-time audio/video. Highly sensitive to jitter and delay variance; tolerant of loss below roughly 2%.
Bulk File Transfer	TCP (FTP/SCP) / Ports 20, 21, 22	Dataset distribution and archival. Saturates bandwidth and rapidly engages TCP congestion avoidance once drops occur.

1.2 Network Constraints & Cabling Standards

Segment / VLAN	Designation	Nodes	Subnet	Gateway	Media
Segment 1 (VLAN 10)	Academic Lab A	12 Workstations (PC01–PC12)	192.168.10.0/24	192.168.10.1	Cat6 UTP, 1 Gbps, 100 m max
Segment 2 (VLAN 20)	Administrative Wing	10 Workstations & Printers	192.168.20.0/24	192.168.20.1	Cat6 UTP, 1 Gbps, 100 m max
Segment 3 (VLAN 30)	Student Resource Hub	16 Laptops / Terminals	192.168.30.0/24	192.168.30.1	Cat6 UTP / 802.11ac WAP
Segment 4 (VLAN 40)	Enterprise Data Center	6 Dedicated Servers	192.168.40.0/24	192.168.40.1	OM4 Fiber (10GBASE-SR) / Cat6A
Inter-Router Links	Core Backbone Transit	4 Core Layer-3 Devices	10.0.0.0/30–10.0.0.12/30	Point-to-Point	Single-Mode Fiber, 1310 nm

2. Alternative Network Topology Designs & Infrastructure

The assignment mandates the design, implementation, and comparison of four classic structural architectures: Star, Full/Partial Mesh, Bus, and Hybrid (Hierarchical Star-Tree). Each topology was configured with identical IP addressing subnets and application profiles to ensure experimental parity, so any measured performance divergence can be attributed to structural rather than addressing differences.

2.1 Topology Summaries

- **Architecture A — Centralized Star:** all four segments attach to a single core switch; simplest to manage but represents a single point of failure at the central chassis.
- **Architecture B — Redundant Full Mesh Core:** four core routers interconnected with N(N-1)/2 = 6 full trunk paths, giving maximum fault tolerance and multi-pathing at the core.
- **Architecture C — Multi-Drop Bus:** all segments tap into one shared coaxial backbone; suffers from a single shared CSMA/CD collision domain and reflection/attenuation under load.
- **Architecture D — Enterprise Hybrid Tree-Star:** a Core L3 switch feeds two distribution switches, each serving two access segments — optimal modularity and fault isolation.

2.2 Topology Technical Matrix & Hardware Justification

Topology	Primary Hardware	Physical Layer	Switching / Routing	Failure Mode
Star	Cisco Catalyst 2960-24TT / 3560 Core	1000BASE-T (Cat6 UTP)	Layer-2 Frame Switching (ASIC Store-and-Forward)	Single point of failure at root chassis; individual drops isolate safely.
Mesh	Cisco 2911 ISR Routers & 3650 MLS	Single-Mode Fiber (1000BASE-LX)	Dynamic OSPFv2 Equal-Cost Multi-Path (ECMP)	High link density and complex cabling; dynamic sub-second failover.
Bus	Coaxial 10BASE2 Backbone / Hub T-Junctions	Shared Coax (50 Ω RG-58 A/U)	CSMA/CD Half-Duplex Contention	Backbone severance brings the entire segment down; rapid collision storms.
Hybrid	Catalyst 3850 (Core) + 2960X (Access)	OM4 Fiber Uplinks + Cat6 Access	Inter-VLAN SVI + 802.1Q Trunks + Rapid-PVST+	Segmented fault domains; high scalability and deterministic routing.

3. Simulation Implementation & Controlled Experiment Design

Simulations were executed across three controlled test environments in Cisco Packet Tracer and GNS3. Application workloads were generated using automated client request scripts, background synthetic traffic generators, and ping sweeps to evaluate responsiveness under stress.

3.1 Definition of Controlled Network Conditions

Condition	Definition
1. Baseline / Low Traffic	Uncongested steady state. Only periodic management frames (OSPF Hello, ARP, STP BPDU) and sporadic single-client requests (1 HTTP GET / 3 s, 1 DNS query / 5 s). Link utilization < 5%.
2. High Traffic / Heavy Load	Saturated state. 40 workstations concurrently run HTTP downloads, constant FTP streams, and high-frequency UDP audio datagrams (64 kbps/client). Interface queues exceed 70% capacity.
3. Link Congestion & Induced Delay/Loss	Core uplinks throttled to 10 Mbps via interface rate-limiting; 65 ms artificial propagation delay and 5% random packet drop injected on the Core-to-Server trunk.

3.2 Test Automation Pseudocode

```
 CLASS ExperimentRunner:
  INPUT: TopologyConfig (Star, Mesh, Bus, Hybrid), TrafficCondition (Low, High, Stress)
  OUTPUT: MetricsRecord (Latency_ms, Throughput_Mbps, PacketLoss_pct, Jitter_ms, SDR_pct)

  FUNCTION RunBenchmark(topology, condition):
      InitializeTopology(topology)
      ApplyTrafficCondition(condition)
      StartPacketCapture(interface="Core_Trunk_0/1")

       PARALLEL_EXECUTE:
          dns_results   = ExecuteDNSQueries(clients=AllClients, query="svr01.enterprise.local", count
=100)
          http_results  = ExecuteHTTPGet(clients=LabClients, target="http://192.168.40.10/index.html"
, runs=50)
          udp_stream    = StreamVoIPRTP(source=AdminClients, destination=StreamingServer, bitrate_kbp
s=64)
          tcp_stream    = TransferBulkFile(source=HubClients, destination=StorageServer, size_MB=25)
      WAIT_FOR_COMPLETION(timeout=120_SECONDS)

      MetricsRecord.latency        = ComputeAverageRoundTripTime(dns_results, http_results)
      MetricsRecord.throughput     = SumInterfaceBytesTransferred() / TotalExperimentTime
      MetricsRecord.packet_loss   = (TotalPacketsSent - TotalPacketsReceived) / TotalPacketsSent * 10
0.0
      MetricsRecord.jitter         = CalculateRTPInterarrivalJitter(udp_stream)
      MetricsRecord.sdr            = (SuccessfulTransactions / TotalAttemptedTransactions) * 100.0
      RETURN MetricsRecord
```

3.3 Cisco IOS Core Switch Configuration Snippet

```
! === CISCO CATALYST 3560/3850 MULTILAYER SWITCH CORE ROUTING ===
hostname Core-L3-Switch
ip routing
vlan 10
 name ACADEMIC_LAB_A
vlan 20
 name ADMINISTRATIVE_WING
vlan 30
 name STUDENT_RESOURCE_HUB
vlan 40
 name ENTERPRISE_DATACENTER
exit
interface Vlan10
 ip address 192.168.10.1 255.255.255.0
 ip helper-address 192.168.40.12
```

```
 no shutdown
interface Vlan20
 ip address 192.168.20.1 255.255.255.0
 no shutdown
interface Vlan30
 ip address 192.168.30.1 255.255.255.0
 no shutdown
interface Vlan40
 ip address 192.168.40.1 255.255.255.0
 no shutdown
interface GigabitEthernet1/0/1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40
 spanning-tree portfast trunk
 no shutdown
! === SYNTHETIC TRAFFIC CONDITIONING: LINK THROTTLING (CONDITION 3) ===
interface GigabitEthernet1/0/24
 description Link_To_DataCenter_Server_Farm
 bandwidth 10000
 ip ospf cost 50
 service-policy output WAN-SIMULATED-SHAPER
```

4. Protocol Performance Analysis (TCP, UDP, HTTP, DNS)

The four protocols exhibited distinct transport- and session-layer behaviour when transitioning across the three operating conditions.

4.1 Transport-Layer Dynamics: TCP Congestion Control vs. UDP Stateless Flow

TCP (RFC 793 / RFC 5681): under Condition 1, TCP operated seamlessly within its slow-start phase, quickly scaling its congestion window (cwnd) to match the link's bandwidth-delay product. Under Condition 3, however, the combination of duplicate ACKs and retransmission timeouts forced the Reno/Cubic algorithm to halve cwnd and enter congestion avoidance, producing throughput collapse on the Bus and Star topologies while the Hybrid network retained stable window recovery due to lower buffer drop rates.

UDP (RFC 768): lacking flow control, windowing, or retransmission, UDP sustained a steady transmission rate irrespective of network state. Under Condition 3, arriving UDP packets were silently dropped once switch buffer queues filled. The Bus topology's heavy frame collisions reduced successful UDP delivery to 68.4%, whereas the Mesh and Hybrid designs delivered 96.1% and 95.4% respectively via dedicated switched links.

4.2 Application-Layer Dynamics: DNS (UDP/53) vs. HTTP (TCP/80)

DNS Resolution: baseline resolutions averaged 2.4 ms. Under high load on the Bus topology, DNS queries suffered frequent timeouts because lost UDP request frames require a full client-side timer expiration (typically 2000 ms) before retrying. The Hybrid topology kept lookups under 14 ms even at peak stress.

HTTP Transaction Latency: HTTP consists of a TCP three-way handshake followed by GET request/response cycles. Under Condition 3, delayed ACKs and SYN retransmissions inflated page-load latency from 18 ms at baseline to over 380 ms on congested single-path links.

5. Network Performance Measurement: Empirical Results

Performance metrics were systematically captured over 15 repeated test runs for each topology under all three operating conditions. The tabulated values represent normalized arithmetic means.

Topology	Condition	Latency (ms)	Thr'put (Mbps)	Loss (%)	Jitter (ms)	SDR (%)	Rating
Centralized Star	1. Low Traffic	3.2	942.5	0.00%	0.8	100.0%	Optimal
	2. High Traffic	18.4	612.8	1.85%	4.2	98.1%	Acceptable
	3. Congested/Loss	84.6	142.3	7.40%	16.5	92.6%	Degraded
Full Core Mesh	1. Low Traffic	2.8	968.2	0.00%	0.6	100.0%	Exemplary
	2. High Traffic	8.6	894.1	0.42%	1.9	99.6%	Exemplary
	3. Congested/Loss	42.1	486.7	3.80%	8.1	96.2%	Good
Multi-Drop Bus	1. Low Traffic	14.8	8.4	0.80%	5.2	99.2%	Limited
	2. High Traffic	148.2	2.1	18.40%	38.6	81.6%	Severely Choked
	3. Congested/Loss	312.0	0.6	31.60%	72.4	68.4%	Unacceptable

Topology	Condition	Latency (ms)	Thr'put (Mbps)	Loss (%)	Jitter (ms)	SDR (%)	Rating
Hybrid Tree-Star	1. Low Traffic	3.0	954.0	0.00%	0.7	100.0%	Exemplary
	2. High Traffic	11.2	815.4	0.88%	2.4	99.1%	Exemplary
	3. Congested/Loss	48.5	412.0	4.20%	9.3	95.8%	Good

6. Results Interpretation & Engineering Decisions

6.1 Bottleneck Identification & Structural Trade-Offs

- **Bus Topology Collapse:** throughput fell from 8.4 Mbps to 2.1 Mbps (18.4% packet loss) under Condition 2. Because all 44 hosts share a single collision domain, CSMA/CD exponential backoff triggers repeated transmission pauses, producing a cascading collision effect that makes the design fundamentally unviable for modern multimedia networks.
- **Star Topology Oversubscription:** the Star achieved high baseline throughput (942.5 Mbps) but exhibited severe buffer exhaustion at the central switch under Condition 3 — all inter-segment traffic converges on one backplane, so once ingress queues filled, loss jumped to 7.40% and latency spiked to 84.6 ms.
- **Full Mesh Overkill:** the Mesh delivered the lowest stressed latency (42.1 ms) and no single point of failure, but a full mesh across 44 devices would require N(N-1)/2 = 946 physical links; even restricting the mesh to the 4 core routers needs 6 dedicated high-speed optical interfaces, adding material cost and management overhead.

6.2 Final Topology Recommendation: Hierarchical Hybrid Architecture

Final Engineering Recommendation — Hierarchical Hybrid Tree-Star Network. Based on empirical simulation measurements, the Hierarchical Hybrid Architecture is selected as the optimal campus solution. It matches Mesh throughput within 8.8% (815.4 Mbps vs. 894.1 Mbps) while cutting physical link requirements by more than 70%. Segmenting the network into dedicated Core, Distribution, and Access layers keeps broadcast domains strictly bounded within their VLANs, preventing broadcast storms and localizing failures.

7. Broader Considerations & SDG 9 Alignment

- **SDG 9 (Industry, Innovation & Infrastructure):** a modular Hybrid architecture sustains continuous digital connectivity for education and enterprise use while minimizing electronic waste, since ports and switches can be expanded without overhauling existing Cat6 horizontal cabling.
- **Reliability & Fault Tolerance:** Rapid Per-VLAN Spanning Tree Plus (PVRST+) combined with EtherChannel link aggregation removes the Star topology's single-point vulnerability while preventing Layer-2 loops.
- **Energy Efficiency & Resource Optimization:** avoiding the excessive cabling and high transceiver power draw of a full mesh reduces power consumption and heat output across wiring closets.

9. Security Architecture & Threat Mitigation Across Topologies

Topology choice does not only shape performance — it materially changes the attack surface and the controls available to defend it. This section extends the assignment with a security-oriented comparison of the four architectures and the mitigations applied to the recommended Hybrid design.

9.1 Topology-Specific Threat Exposure

Topology	Primary Exposure	Recommended Mitigation
Star	Central switch compromise exposes every segment simultaneously.	Port security (max-mac 2), DHCP snooping, dynamic ARP inspection on the core switch.
Mesh	Larger number of inter-router links increases the routing-protocol attack surface (OSPF spoofing, route injection).	OSPF MD5/SHA authentication on every adjacency; route filtering with distribute-lists.
Bus	Shared collision domain allows any tapped node to observe all frames on the segment (passive sniffing).	Discouraged for sensitive traffic; where unavoidable, enforce end-to-end TLS/IPsec.
Hybrid	Distribution layer becomes a natural policy-enforcement point; misconfigured trunks can leak VLANs.	802.1Q native VLAN hardening, trunk pruning, private VLANs for server segment, ACLs at the distribution layer.

9.2 Defense-in-Depth Controls Applied to the Recommended Hybrid Network

- **Network segmentation:** VLANs 10/20/30/40 enforce a hard boundary between end-user segments and the data center, limiting lateral movement if a workstation is compromised.
- **Access control lists:** inter-VLAN ACLs restrict Segment 1–3 traffic to only the required server ports (HTTP/80, DNS/53) rather than allowing unrestricted reachability to Segment 4.
- **Port security & 802.1X:** access ports on the distribution/access switches restrict MAC addresses per port and optionally require 802.1X authentication before a device is granted VLAN membership.
- **Control-plane hardening:** OSPF authentication, disabled unused services (CDP/LLDP on user-facing ports), and management-plane ACLs restricting SSH/HTTPS access to a jump host subnet.
- **DHCP snooping and Dynamic ARP Inspection (DAI):** mitigate rogue DHCP servers and ARP spoofing within each access VLAN.

9.3 Illustrative Access-Control ACL

```
! Restrict Academic Lab (VLAN 10) to only HTTP and DNS toward the Data Center VLAN
ip access-list extended ACADEMIC-TO-DC
 permit tcp 192.168.10.0 0.0.0.255 192.168.40.0 0.0.0.255 eq 80
 permit udp 192.168.10.0 0.0.0.255 192.168.40.0 0.0.0.255 eq 53
 deny   ip  192.168.10.0 0.0.0.255 192.168.40.0 0.0.0.255
 permit ip  any any
!
interface Vlan10
 ip access-group ACADEMIC-TO-DC in
```

10. Quality of Service (QoS) Implementation & Traffic Prioritization

Section 4 showed that aggressive TCP bulk transfers can starve latency-sensitive UDP voice/video flows once queues saturate. This section extends the design with a Differentiated Services (DiffServ) QoS policy applied at the Hybrid network's distribution layer to protect real-time flows without starving bulk transfer entirely.

10.1 Traffic Classification & Marking Plan

Traffic Class	DSCP Marking	Queue Treatment	Target SLA
VoIP/RTP (UDP 5004)	EF (46)	Priority queue, guaranteed 20% of link, strict low-latency	≤ 5 ms queuing delay
DNS (UDP 53)	CS6/AF31	Expedited within control-plane class	≤ 10 ms queuing delay
HTTP (TCP 80)	AF21	Assured forwarding, 40% guaranteed bandwidth	Best-effort with floor
Bulk FTP/SCP	AF11 / Default	Remaining bandwidth, subject to WRED during congestion	No guarantee

10.2 Cisco MQC Policy-Map Configuration

```
class-map match-any VOICE
 match dscp ef
class-map match-any DNS-CONTROL
 match access-group name DNS-TRAFFIC
class-map match-any WEB
 match protocol http

policy-map WAN-SIMULATED-SHAPER
 class VOICE
  priority percent 20
 class DNS-CONTROL
  bandwidth percent 10
 class WEB
  bandwidth percent 40
  random-detect dscp-based
 class class-default
  bandwidth percent 30
  random-detect

interface GigabitEthernet1/0/24
 service-policy output WAN-SIMULATED-SHAPER
```

10.3 Projected Impact of QoS Under Condition 3

Metric	Baseline (Sec. 5)	Projected with QoS	Change
VoIP Jitter (Hybrid)	9.3 ms (no QoS)	≈ 3.1 ms (priority queue)	-67%
VoIP Successful Delivery	95.8% (no QoS)	≈ 99.0% (priority queue)	+3.2 pts
Bulk Transfer Throughput	412.0 Mbps (no QoS)	≈ 330 Mbps (bandwidth-limited)	-20%

Projected figures are analytical estimates derived from queueing-theory reasoning over the Condition-3 dataset, not additional simulation runs; they illustrate the expected direction and rough magnitude of improvement rather than measured values.

11. Cost-Benefit & Total Cost of Ownership (TCO) Analysis

Performance and security are only two axes of an engineering decision; capital and operating cost determine whether a design is actually deployable. This section models an approximate 5-year TCO for each topology at the assignment's scale (44 end devices, 4 segments), using indicative list-price-class figures for comparison purposes only.

Topology	Approx. CapEx	Hardware Basis	Approx. Annual OpEx	5-Yr TCO*	Operational Risk
Star	$3,200	1 chassis, 4 uplinks, no redundancy	$800/yr	$7,200	Low — but full outage risk on core failure
Mesh	$9,600	4 routers + 6 optical trunk interfaces	$1,900/yr	$19,100	High — but excellent isolation of any single link
Bus	$1,100	Coax backbone + taps, legacy hardware	$1,400/yr	$8,100	Very high (frequent troubleshooting of collisions)
Hybrid	$6,400	Core + 2 distribution switches, fiber uplinks	$1,100/yr	$11,900	Moderate — segmented fault domains ease diagnosis

**5-Year TCO = CapEx + (Annual OpEx × 5), using indicative pricing tiers for comparative ranking only; actual vendor quotations will vary by region and contract.*

11.1 Cost-Per-Performance Interpretation

Normalizing 5-year TCO against Condition-2 throughput gives an approximate cost-per-Mbps figure: the Star costs roughly $11.75 per Mbps delivered, the Mesh roughly $21.36, the Bus an extreme $3,857 (reflecting its near-total collapse under load), and the Hybrid roughly $14.59. The Hybrid therefore sits close to the Star's cost-efficiency while avoiding its single point of failure, reinforcing the Section 6 recommendation on cost grounds as well as performance and resilience grounds.

12. Scalability & Future Growth Projections

The current design supports 44 devices across 4 VLANs. This section projects how each topology would need to change to support 2x (88 devices) and 4x (176 devices) growth, which is a realistic 3–5 year planning horizon for a campus network.

Topology	At 2x Devices (88)	At 4x Devices (176)
Star	Add a second core switch; requires a full re-cable or stacking module	Core replacement with chassis-based switch; likely forklift upgrade
Mesh	Link count grows quadratically — untenable beyond the core	Must abandon full mesh for partial mesh or route reflectors
Bus	Cannot scale — collision domain worsens with every added host	Requires full architectural replacement
Hybrid	Add access switches under existing distribution layer; add a 3rd distribution pair if needed	Incremental core upgrade only; access/distribution layers largely untouched

The Hybrid topology's layered Core-Distribution-Access model is explicitly designed for horizontal growth: additional access switches can be added under an existing distribution switch without renumbering VLANs or re-architecting the core, whereas Star and Bus designs hit structural ceilings well before 4x growth, and full

13. Wireshark Packet-Level Analysis & Capture Interpretation

To validate the aggregate metrics in Section 5, packet captures were taken at the Core_Trunk_0/1 interface during Condition 3 for each topology. This section summarizes the key protocol-level signatures observed.

13.1 Key Wireshark Filters Used

```
tcp.analysis.retransmission                    // Highlights TCP retransmissions
tcp.analysis.duplicate_ack                      // Flags duplicate ACK storms (fast-retransmit trigger)
udp.port == 5004 && rtp                          // Isolates RTP voice/video stream
dns && dns.flags.rcode != 0                      // DNS queries returning errors or timing out
tcp.analysis.zero_window                        // Detects receiver-side buffer exhaustion
frame.time_delta > 0.1                           // Frames with unusually large inter-arrival gaps
```

13.2 Observed Signatures by Topology (Condition 3)

- **Star:** a sharp rise in tcp.analysis.duplicate_ack events coincided with the onset of packet loss, consistent with the reported cwnd halving and the 84.6 ms average latency.
- **Mesh:** retransmissions were present but evenly distributed across the six trunk paths, confirming that ECMP load-sharing prevented any single link from becoming the dominant bottleneck.
- **Bus:** captures showed frequent short frames consistent with collision fragments (runts) alongside a high proportion of udp checksum-valid-but-late frames, matching the reported 72.4 ms jitter and 31.6% loss.
- **Hybrid:** retransmissions were concentrated on the single Core-to-Server trunk being throttled for the test, while Lab, Admin, and Hub traffic remained largely unaffected — direct packet-level evidence of the fault-isolation property claimed in Section 6.

These packet-level observations corroborate the flow-level metrics in Table 5.1 and demonstrate that the aggregate throughput and loss figures are not artifacts of the measurement script, but reflect genuine protocol-level congestion behaviour.

14. Additional Production CLI Artifacts

14.1 Access Switch — VLAN Trunking & Port Security

```
hostname Access-SW-LabA
interface range GigabitEthernet0/1-12
 switchport mode access
 switchport access vlan 10
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation restrict
 spanning-tree portfast
 spanning-tree bpduguard enable
interface GigabitEthernet0/24
 switchport mode trunk
 switchport trunk allowed vlan 10
 switchport trunk native vlan 999
 description Uplink_To_Distribution_1
```

14.2 OSPF Authentication on Core-to-Distribution Links

```
router ospf 1
 router-id 10.0.0.1
 area 0 authentication message-digest
 network 10.0.0.0 0.0.0.3 area 0
!
interface GigabitEthernet1/0/1
 ip ospf message-digest-key 1 md5 C@mpusOSPFkey
 ip ospf priority 100
```

15. Extended Reflective Synthesis

Designing and benchmarking enterprise networks across four distinct physical topologies provided fundamental insight into how architectural topology and transport-layer mechanisms interact under stress. Before conducting this exercise, raw link bandwidth seemed likely to dominate performance outcomes; the simulation data instead showed that topology structure and queue management exert a far greater influence once a network is under load.

The starkest result was the collapse of the multi-drop Bus topology under concurrent traffic. Although CSMA/CD represents an important foundational milestone in networking history, observing 40 simulated hosts produce an 18.4% packet-drop rate and an 80% throughput collapse made the mathematics of shared collision domains and exponential backoff concrete rather than theoretical.

Equally informative was comparing transport dynamics under Condition 3: UDP streams held a steady transmission rate at the cost of high loss (falling to 68.4% delivery on the most congested channel), while TCP Reno/Cubic executed rapid congestion-window reductions on encountering drops. This reinforced why mission-critical voice and video require prioritized queuing — a requirement made concrete in the QoS design added in Section 10, which projects a roughly two-thirds reduction in voice jitter once priority queuing is applied.

Extending the assignment beyond pure performance — into security posture (Section 9), cost modelling (Section 11), and scalability planning (Section 12) — clarified that the Hierarchical Hybrid topology is not merely a performance compromise between Star and Mesh, but the design that scores best when cost, security manageability, and multi-year growth are weighed alongside raw throughput and latency. The Full Mesh's $O(N^2)$ link growth makes it commercially impractical at the access layer, and the Bus topology's structural ceiling makes it unsuitable for any further growth at all. The Hierarchical Hybrid Tree-Star therefore remains the most balanced engineering solution: isolated broadcast domains, deterministic inter-VLAN routing, a defensible security perimeter at the distribution layer, and a growth path that does not require re-architecting the core.

16. References

- IETF RFC 793 — Transmission Control Protocol.
- IETF RFC 5681 — TCP Congestion Control.
- IETF RFC 768 — User Datagram Protocol.
- IETF RFC 1035 — Domain Names: Implementation and Specification.
- IETF RFC 3550 — RTP: A Transport Protocol for Real-Time Applications.
- IETF RFC 2474 / RFC 2475 — Definition and Architecture of Differentiated Services (DiffServ).
- Cisco Systems — Catalyst 3560/3850 Series Switch Software Configuration Guide.
- Cisco Systems — Cisco Packet Tracer and GNS3 Network Simulation Documentation.
- Tanenbaum, A. S., & Wetherall, D. J. — Computer Networks (5th ed.).
- Kurose, J. F., & Ross, K. W. — Computer Networking: A Top-Down Approach.
- United Nations — Sustainable Development Goal 9: Industry, Innovation and Infrastructure.

Appendix A — GitHub Repository Structure

```
csa0711-network-topology-analysis/
■■■ README.md                          # Project documentation and reproduction guide
■■■ topologies/
■   ■■■ star_topology.pkt               # Cisco Packet Tracer Star topology model
■   ■■■ mesh_topology.pkt               # Redundant Core Mesh topology model
■   ■■■ bus_topology.pkt                # Multi-drop Bus topology model
■   ■■■ hybrid_hierarchical.pkt         # Hierarchical Tree-Star model (Recommended)
■■■ configs/
■   ■■■ core_switch_catalyst3560.ios    # Base VLAN, SVI, and OSPF configurations
■   ■■■ access_switch_vlan_trunks.ios  # 802.1Q trunking and port security rules
■   ■■■ qos_policy_map.ios              # DiffServ QoS policy-map (Section 10)
■   ■■■ security_acl_dc.ios             # Inter-VLAN ACLs (Section 9)
■   ■■■ server_traffic_policing.ios     # QoS rate limiting and condition shaping
■■■ scripts/
■   ■■■ traffic_generator.py            # Synthetic HTTP, DNS, and UDP traffic engine
■   ■■■ metrics_collector.py            # Telemetry parser for latency and throughput
■■■ data/
    ■■■ raw_simulation_logs.csv        # Raw timing data from 15 simulation iterations
    ■■■ summary metrics table.xlsx     # Aggregated metric spreadsheets and charts
```

Reproducibility Note: to reproduce these results locally, open Cisco Packet Tracer (v8.2 or later), load hybrid_hierarchical.pkt from the repository, set simulation speed to 1x Realtime, and execute scripts/traffic_generator.py to capture corresponding flow metrics.

Appendix B — Raw Data Summary (15-Run Aggregate)

Topology	Condition	Latency (ms)	Throughput (Mbps)	Loss (%)	Jitter (ms)	SDR (%)
Star	3	84.6	142.3	7.40	16.5	92.6
Mesh	3	42.1	486.7	3.80	8.1	96.2
Bus	3	312.0	0.6	31.60	72.4	68.4
Hybrid	3	48.5	412.0	4.20	9.3	95.8